

Beyond Graduation: A Deep Dive into Irish Graduate Employment and Earnings



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Statement of Authorship

We, the undersigned, hereby declare that this assignment is our original work and has been entirely prepared in alignment with the academic integrity policies of UCD Michael Smurfit Business School. We confirm that:

1. The statistical analysis was collaboratively developed and completed by our team, based on the provided assignment objectives.
2. The content of this report is our own work, and where external sources have been used, they are properly cited and acknowledged.
3. All team members have contributed equally to the completion of the assignment, and their contributions are detailed in the report.
4. We have not engaged in plagiarism, collusion, or any other form of academic dishonesty in preparing this report.

By signing below, we affirm that this declaration is true and correct to the best of our knowledge:

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Executive Summary

This report explores the employment outcomes, industry trends, and salary dynamics of higher education graduates in Ireland. It analyses the impact of factors such as academic performance, institutional tier, gender, field of study, and years since graduation. Various statistical techniques, including descriptive statistics, two-proportion Z-tests, ANOVA, and confidence intervals, were used to assess these factors.

This report found that in contrast to 83% of 2018 graduates, about 80% of 2019 grads found "substantial employment" within the first year of graduation. Median earnings of 2019 graduates after one year of graduation was found to be €662 per week.

Graduates in fields like ICT and Business experienced notable salary growth, while those in sectors such as Arts and Services saw limited improvement. In the employment sector it is observed that 43% employees are fresh graduates (Years since graduation: 1-2). Most common industries for Male graduate in 2019 was – *Professional, Scientific and Technical Activities* and for females - *Education and Human Health and Social Work Activities*.

This analysis was carried out by students of Business Analytics from UCD Smurfit Business School for the Statistical Methods Group Assignment. The graduation outcomes data studied are fetched from Central Statistics Office.

These findings provide deeper understanding of the opportunities and challenges faced by graduates in Ireland's labour market.



Introduction

Ireland has emerged as a prominent destination for higher education, attracting both domestic and international students. The country's globally recognized universities, strong ties to industry, and vibrant cultural environment contribute significantly to its appeal. Ireland offers high-quality education in fields such as technology, business, and healthcare, which align with global market trends and the growing demand for skilled professionals. Additionally, Ireland's post-study work opportunities, notably the Third Level Graduate Scheme, provide international graduates with a chance to gain valuable work experience, further enhancing the country's attractiveness as an educational hub.

The analysis of graduate outcomes in higher education is vital for understanding the long-term impacts of educational qualifications on employment and earnings. This report focuses on data sourced from the CSO (Central Statistics Office) on higher education outcomes for graduates in Ireland, providing valuable insights into the career paths and earnings of graduates across various fields of study, institutions, and demographic groups. By examining this data, we aim to provide a comprehensive view of how higher education contributes to employment outcomes and how factors such as gender, field of study, grade level, and institution type influence these outcomes.

Our analysis can offer data-driven insights into higher education in Ireland that may be valuable for:

- 1) Prospective students to learn more about the Irish education sector
- 2) Institutions to understand market trends
- 3) Employers to assess their current workforce and use it as a benchmark for future hiring

Our focus will be on five key aspects: the number of graduates and their employment outcomes, the sectors in which they are employed, and earnings information, with a specific focus on gender, field of study, NFQ level, institution, grade, and years since graduation.

Through this analysis, we seek to answer several important questions:

1. Which gender is performing well in securing employment?
2. Is there gender-based salary disparities among graduates?
3. How does the level of education (NFQ level) impact employability and earnings?
4. Are fresh graduates getting employed in the first 2 years of graduation with the minimum salary required for general employment and work permit?
5. Which institutions have the highest employment success rates for their graduates?
6. What is the correlation between field of study and sector of employment?
7. How does grade level influence employment outcomes?
8. How does the salary change over time?

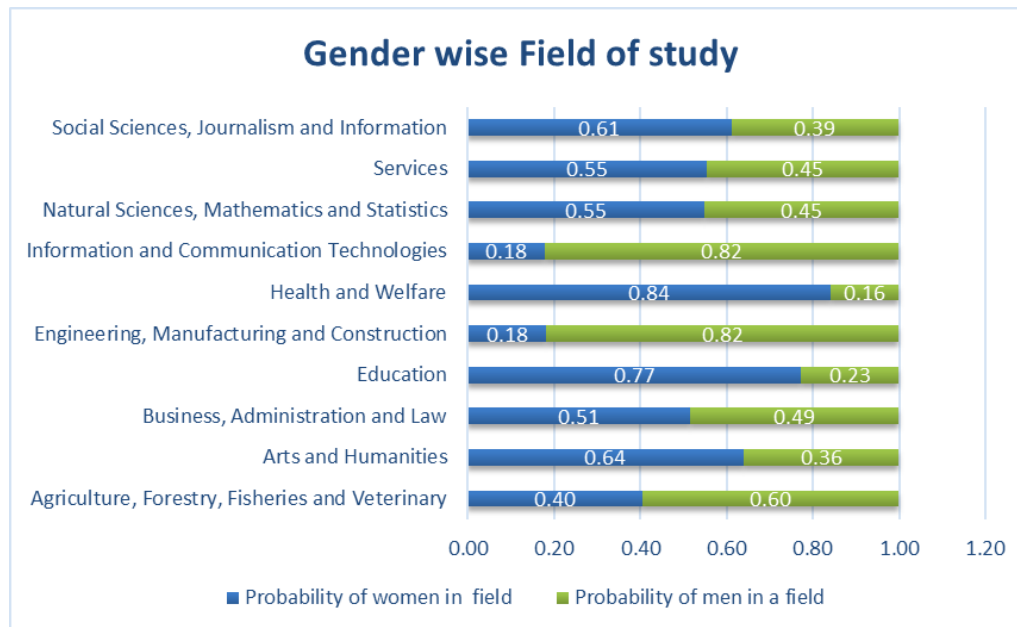
This analysis aims to provide valuable insights for prospective students, institutions, and employers, helping to guide future decisions regarding education and workforce planning.



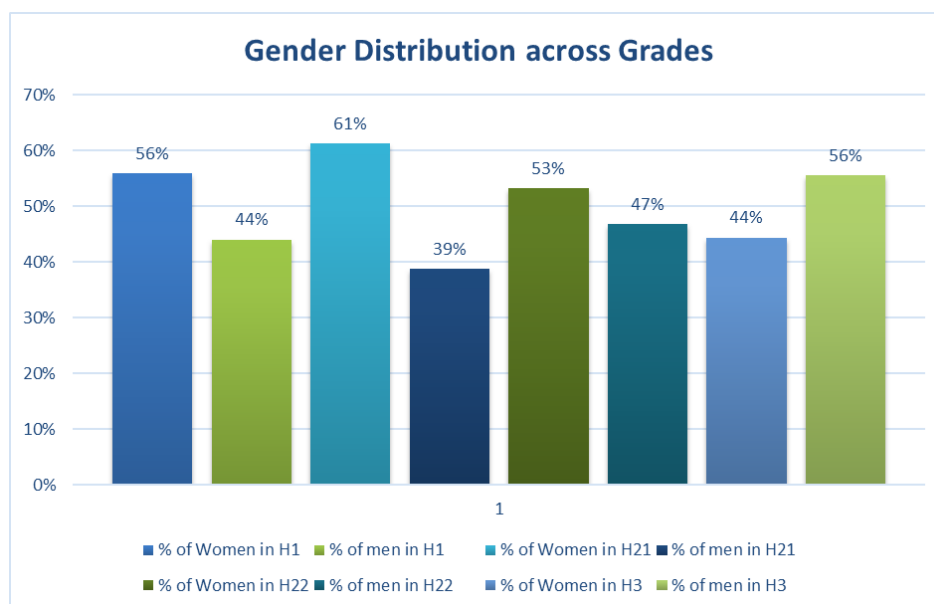
What Do Graduates Study?

This section focuses on various employment outcomes for graduates.

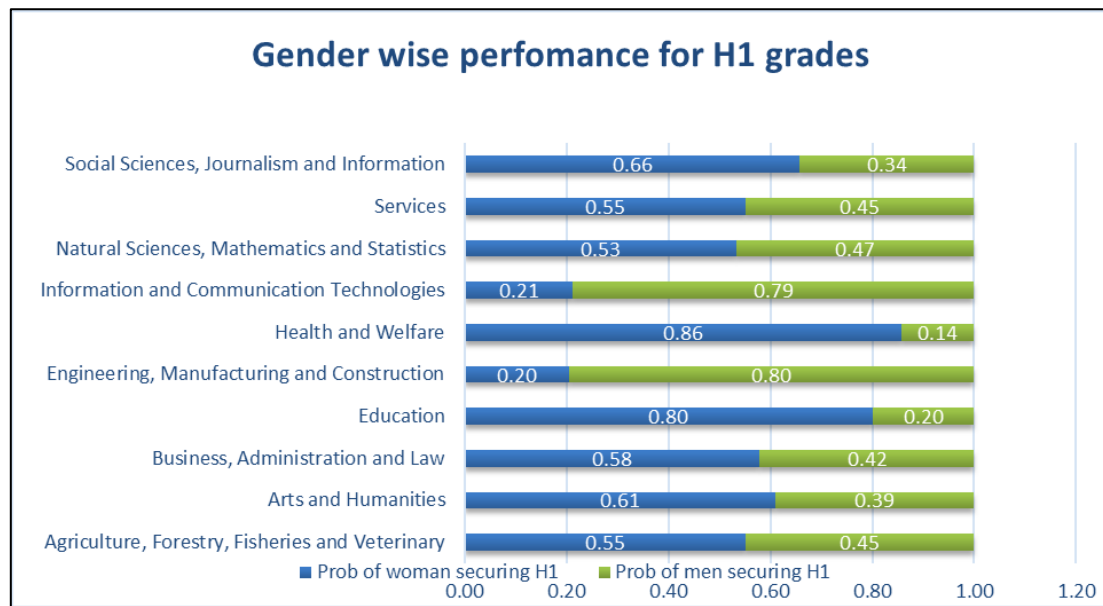
The data reveals gender disparities across fields: women dominate **Health and Welfare (84%)** and **Education (77%)**, while men lead **Engineering, Manufacturing, and ICT (82%)** and **Information and Communication Technology (82%)**. Social Sciences and Services show more balance.



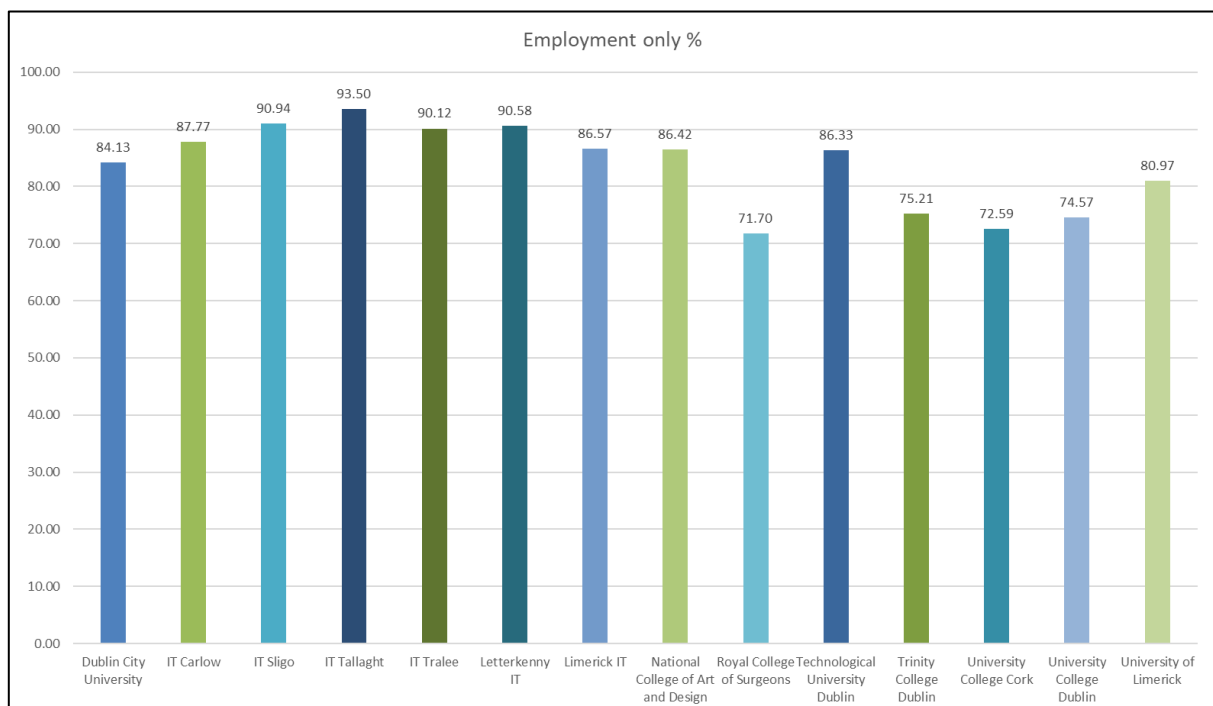
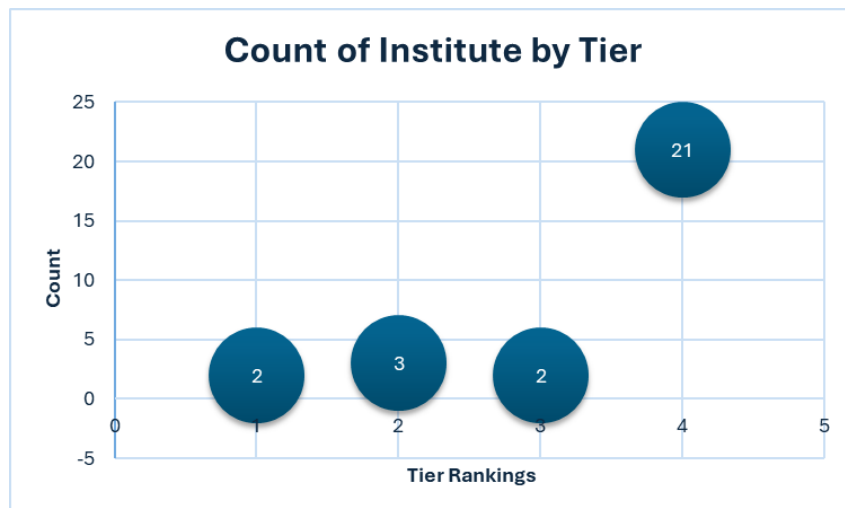
The data shows gender trends in academic performance, with women excelling in higher grades like **H1 (56%)** and **H2.1 (61%)**, while men slightly dominate in lower grades like **H3 (56%)**.



Below graph displays that women perform well in most fields except for IT and Engineering fields.

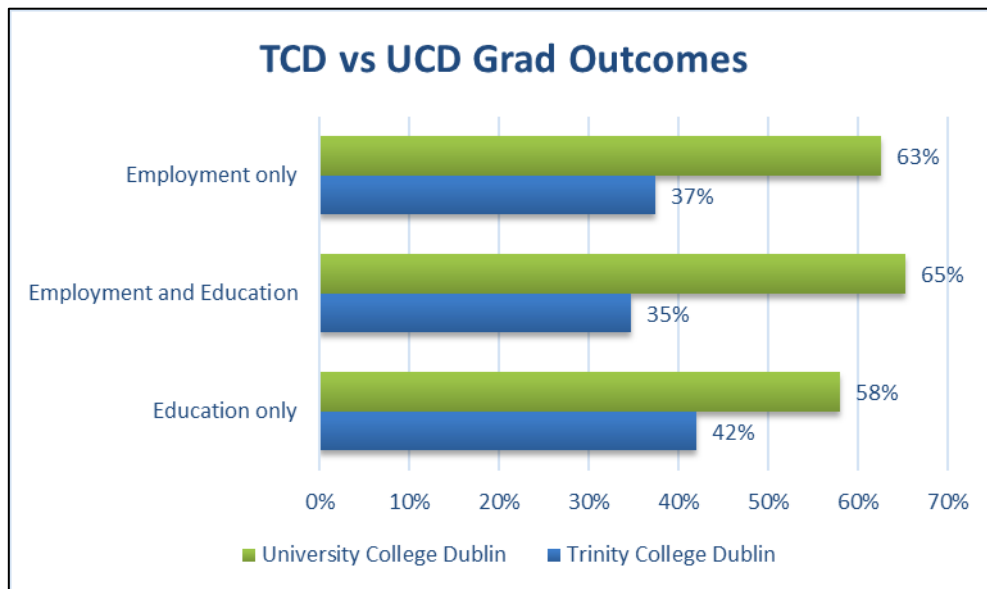


Total of 27 Institutes are divided into four tiers.

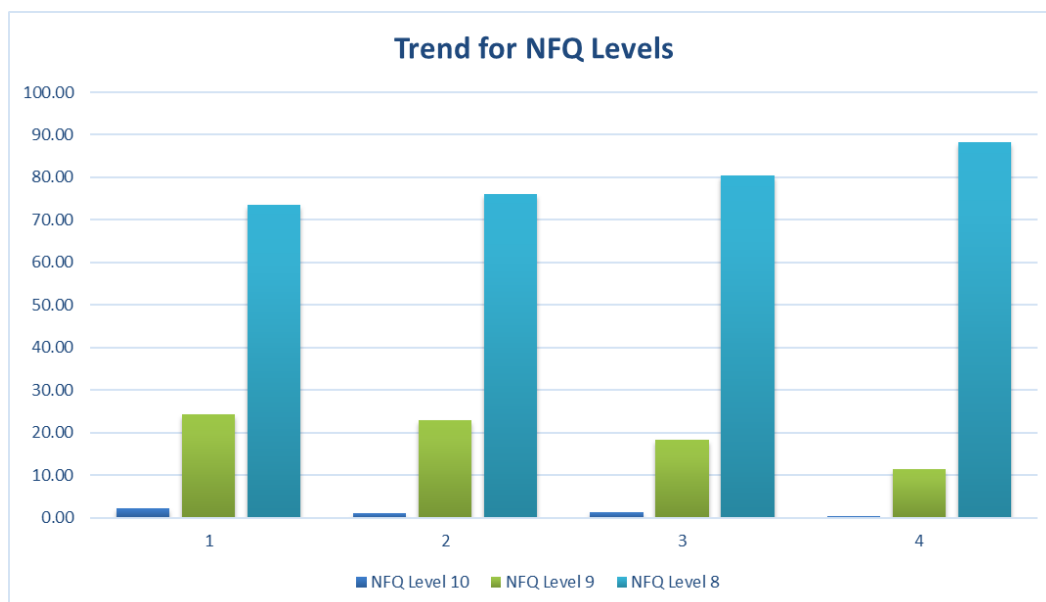


Employment Dominance: Institutions like IT Tallaght (93.50%) and IT Sligo (90.94%) have the highest percentage of graduates transitioning directly to employment, indicating strong industry connections or job placement services.

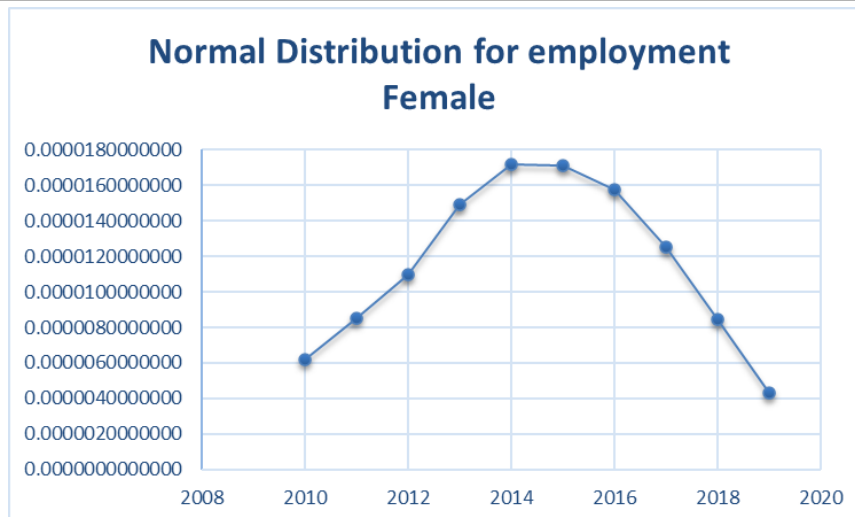
TCD (75.21%) and **UCD (74.57%)** show strong employment placement rates, reflecting high market demand for graduates from these prestigious institutions.



University College Dublin outshines **Trinity College Dublin** in preparing graduates for employment, with **63%** entering the workforce compared to **37%** from Trinity, highlighting UCD's stronger focus on direct job placement.



Below is the normal distribution plot for women in employment, as most of our hypothesis tests assume a normal data distribution.



Inferential Statistics

Confidence Intervals

The sample of women showed a mean employment score of **178**, with a standard deviation of **138.14**, indicating significant variability in the data.

Confidence intervals were calculated at 90%, 95%, and 99% confidence levels:

- **90% CI:** 96.65 to 179.62
- **95% CI:** 88.71 to 187.57
- **99% CI:** 73.17 to 203.10

These intervals estimate the range within which the true population mean likely lies, with wider intervals reflecting higher confidence but less precision.

Impact of Higher Grades on Employment Probability – 2 Proportions Z-Test

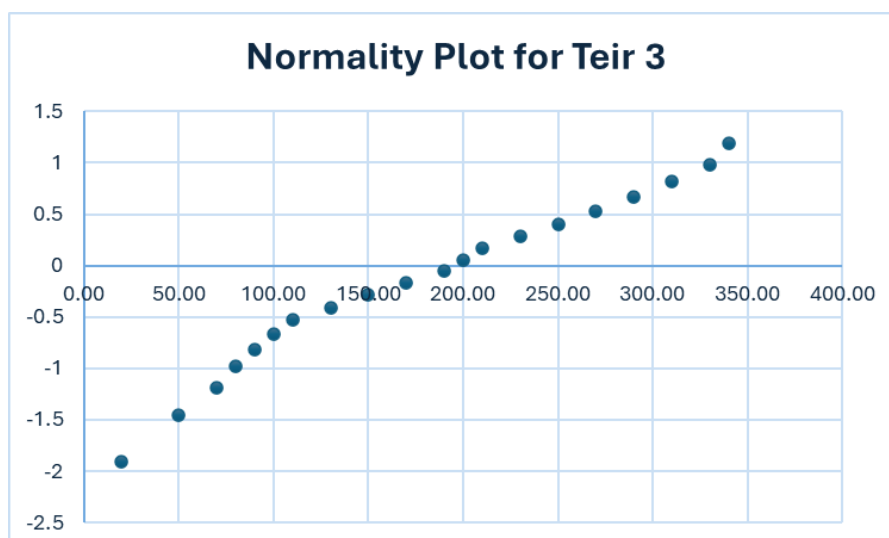
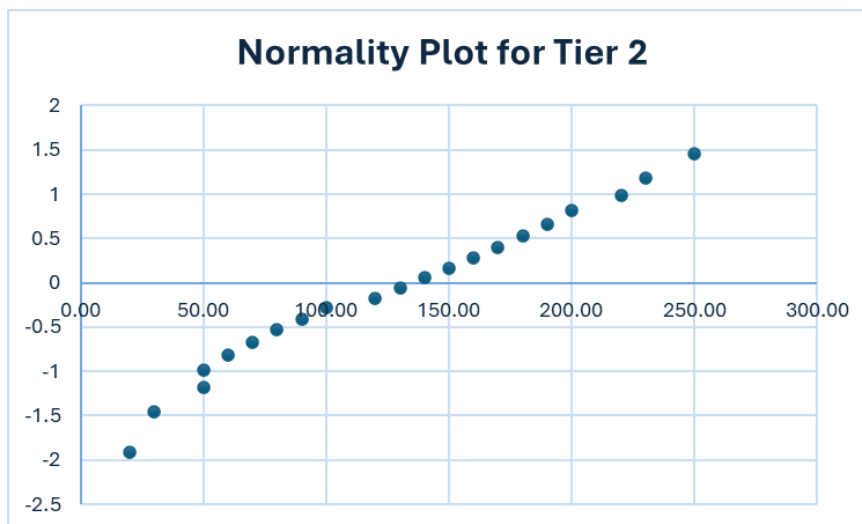
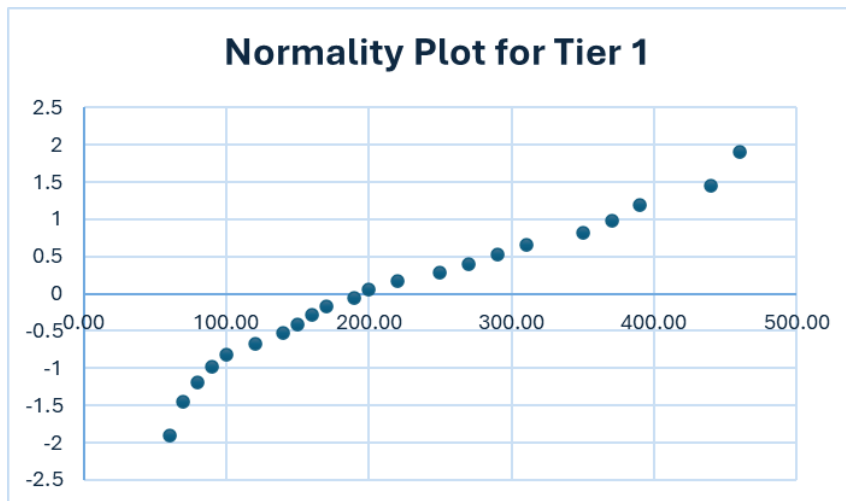
For Female:

The analysis examined if higher grades (H1 and H21) increase employment likelihood for women compared to lower grades (H22 and H3). The hypothesis test aims to determine if the proportion of one group (p_1) exceeds that of another (p_2). The null hypothesis (H_0) assumes no difference ($p_1 - p_2 = 0$), while the alternative hypothesis (H_a) suggests $p_1 - p_2 > 0$, making it a **right-tailed test**.

A z-test revealed a **test statistic** of -0.28 and a **p-value** of 0.610, exceeding the **0.05 significance level**. As the p-value was higher than the significance level, the null hypothesis was not rejected, indicating no significant difference in employment likelihood between the two groups. Therefore, higher grades do not significantly increase the chances of employment for women in this sample. Employers may prioritize soft skills, work experience, or internships over grades in women's employment outcomes. Exploring industry preferences and systemic hiring factors can guide actionable improvements in employability strategies.

A similar hypothesis test for **men** showed a **test statistic** of 0.14 and a **p-value** of 0.443, which exceeds the 0.05 significance level. This means there's insufficient evidence to suggest higher grades significantly affect employment chances for men.

Normality Plot for Anova Test on 3 tiers of Institutes:



The scatter plot of z-scores and data-scores among the 3 Tiers, represents a roughly linear trend, which allows us to assume that the variable is normally / approximately normally distributed.

ANOVA Test on Tier 1, Tier 2, and Institutes Impacting Employment

An ANOVA test was conducted to examine whether there are significant differences in employment outcomes based on tier rankings (Tier 1, Tier 2, Tier 3) in Ireland. The data involved three samples with the **Overall Sample Mean**: 177.14. The null hypothesis, $H_0: \mu_1 = \mu_2 = \mu_3$, was tested against the alternative hypothesis that **at least one population mean is different**. Since the p-value (0.010) is less than the significance level of 0.05, we rejected the null hypothesis.

The analysis yielded the following results:

- **Test Statistic:** 4.77
- **Critical Value:** 3.15
- **p-value:** 0.010
- **Significance Level (α):** 0.05

Conclusion: With 5% significance, the analysis shows that tier and institutional differences significantly affect employment outcomes, with variations across tiers and institutes. This finding highlights disparities in institutional effectiveness in workforce transitions, influenced by resources, connections, or focus, guiding stakeholders to address or leverage differences to enhance graduate employability.

Where do Graduates Work?

This section focuses on the industry classification where graduates are employed. It includes only those graduates who are classified as being in 'substantial employment'. This analysis aims to uncover key patterns within the dataset and derive actionable insights to guide educational institutions and policymakers. The study specifically focuses on average employment outcomes across various fields of study, years since graduation, grade levels, and gender-based differences.

To start with, the grade levels of employed graduates were examined to assess the impact of academic performance on securing employment. The objective was to identify any patterns or correlations between higher grades and more stable job prospects. Interestingly, the analysis revealed insights that challenge the widely held belief that excellent grades are always a prerequisite for better employment opportunities.



The graphical representation shows 48% of the employed graduates are H21 honours (60 - 69% marks) and only 13% (70 - 100% marks) employers comes under H1 Honours.

To explore potential gender disparities in employment, the average employment rates for male and female were calculated. The analysis revealed that 12% more females are employed compared to males.

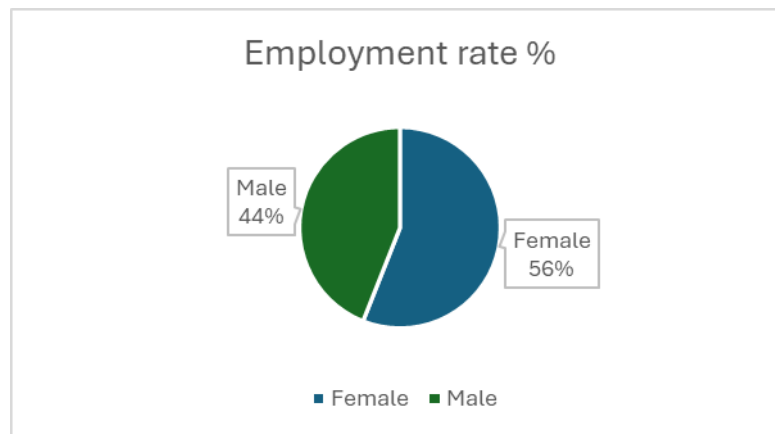


Fig 2

In analysing the field of study, the most popular course among both male and female graduates is found to be **Business, Administration, and Law**. Furthermore, confidence intervals are used to estimate the average number of graduates in the fields of Business, Law, and Administration based on a sample dataset. Using confidence intervals, we assess the range within which the true population mean is expected to fall at varying confidence levels (90%, 95%, and 99%) and at a 95% confidence level, the average number of graduates is estimated to be between 146.75 and 263.25.

Additionally, sector-wise employment trends reveal that **Education** is predominantly a female-dominated field, with women comprising 70% of employees. In contrast, the **Professional, Scientific, and technical** sector is male dominated, with 52% of employees being men.

Inferential Statistics

Two Mean Z-Test

To determine whether the proportion of female employment is significantly greater than male employment among the graduates a right-tailed **Two Mean Z-Test** is performed.

The test statistic value -1.44 falling below critical value at the 5% significance level; the data does not provide sufficient evidence to conclude that the proportion of female graduates employed is greater than that of male graduates.

Upon examining the total database, it was initially observed from descriptive statistics that the proportion of female graduates employed was higher than that of male graduates. This suggested that female graduates had a higher employment rate compared to their male counterparts as already mentioned through the graph on descriptive analytics.

However, when hypothesis testing was performed using a simple random sampling method, the results were contradictory. The sample obtained had a higher proportion of male graduates employed compared to female graduates, which led to a failure to reject the null hypothesis.

One Proportion Z-Test

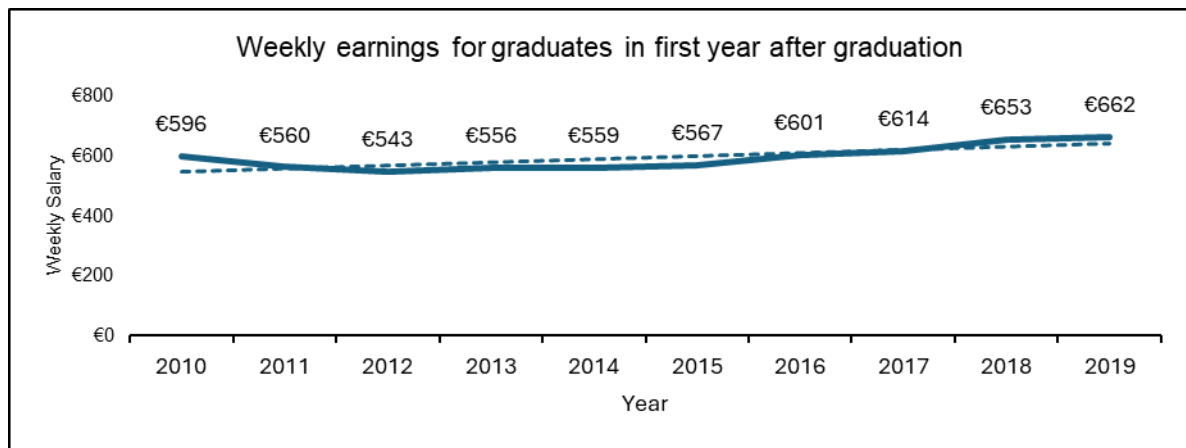
Regarding years since graduation, **One Proportion Z-Test** is performed for Employment Within 2 Years of Graduation. This test aims to evaluate whether more than 50% of graduates in Ireland are employed within two years of graduation using a right-tailed test.

The test statistic -6.39 falling below critical value indicates that at the 5% significance level, the data do not provide sufficient evidence to conclude that more than 50% of graduates in Ireland are employed within two years of graduation. This finding implies that, based on this sample, out of the graduates who got employed in 10 years, less than 50% only secured a job in the first two years.

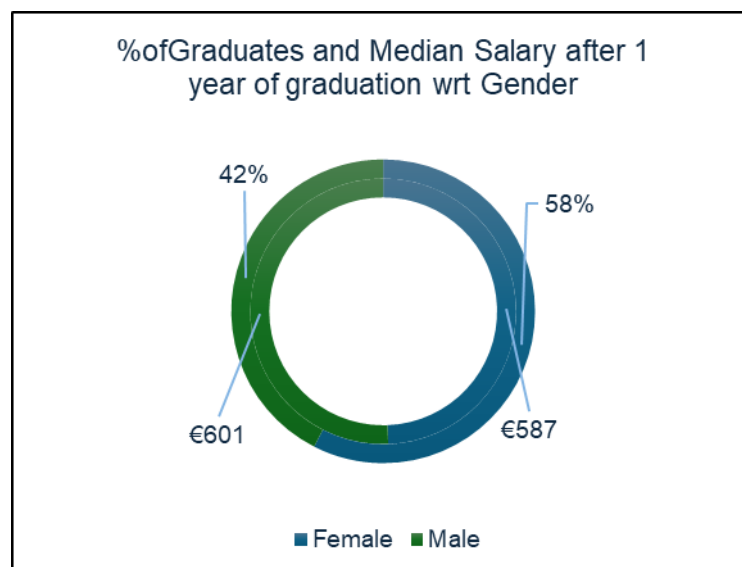
What do Graduates Earn?

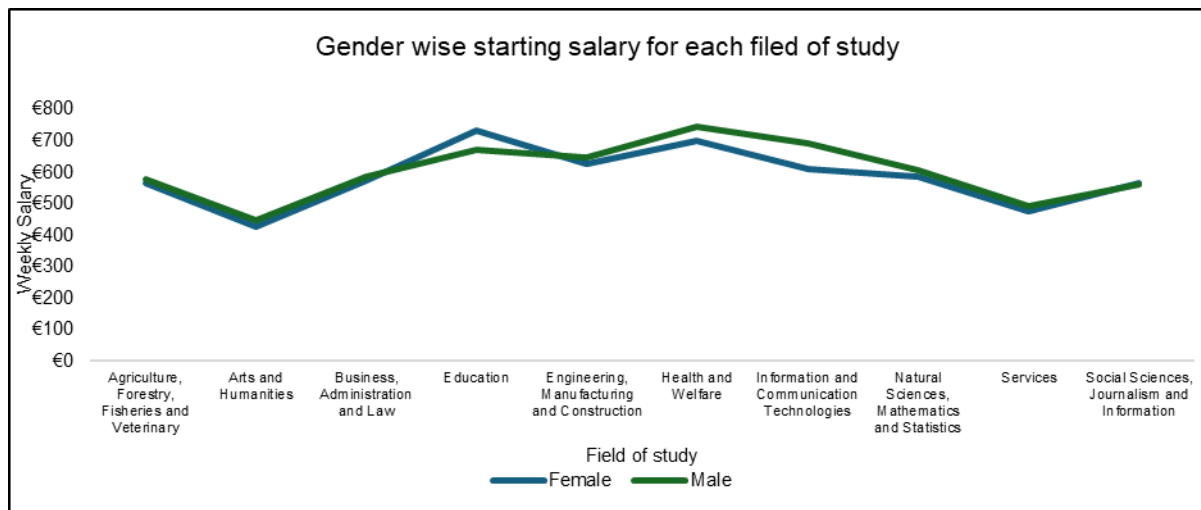
This section discusses in-depth about the weekly salary outcomes for higher education graduates in Ireland. Salary data is available as median values of earning rounded to the nearest €5. Descriptive and inferential statistical techniques were employed to analyse trends across parameters such as gender, field of study, NFQ levels, and institutional tiers. The analysis primarily focuses on either starting salaries (within one year of graduation) or the salary trends for 2010 graduates over the subsequent ten years. Due to data unavailability and time constraints, given median salaries were assumed to represent. Consequently, the findings and insights should be interpreted with this limitation in mind.

Descriptive Statistics

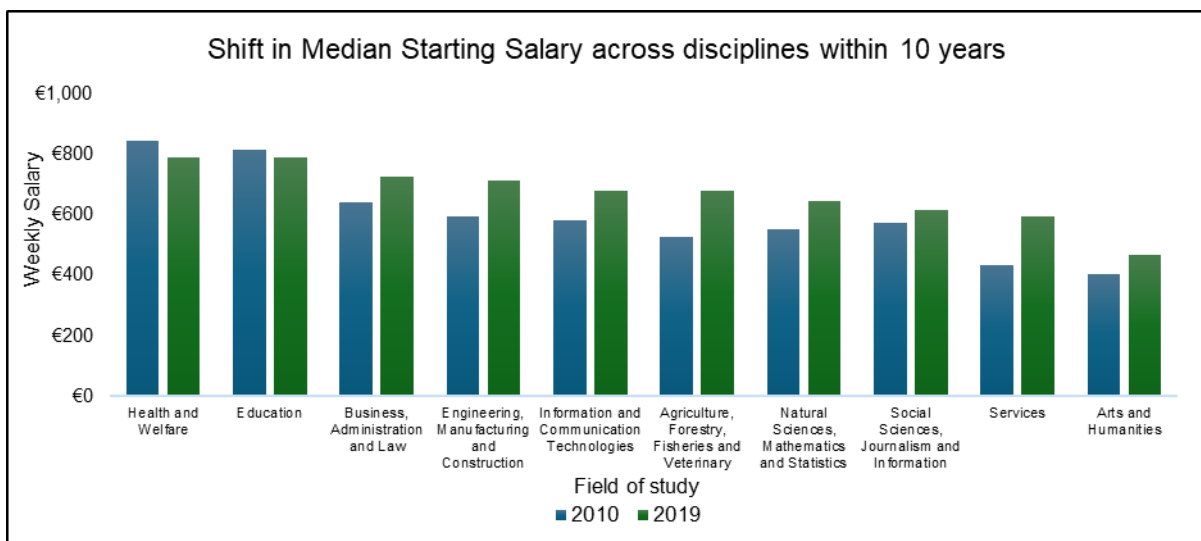


The graph shows that mean **weekly salary increased by 11% over a decade**, from €596 to €662. A 95% confidence interval places the mean salary within €528 and €582, highlighting stable growth despite economic fluctuations. However, the modest pace of salary growth highlights a need for strategies to accelerate economic returns.



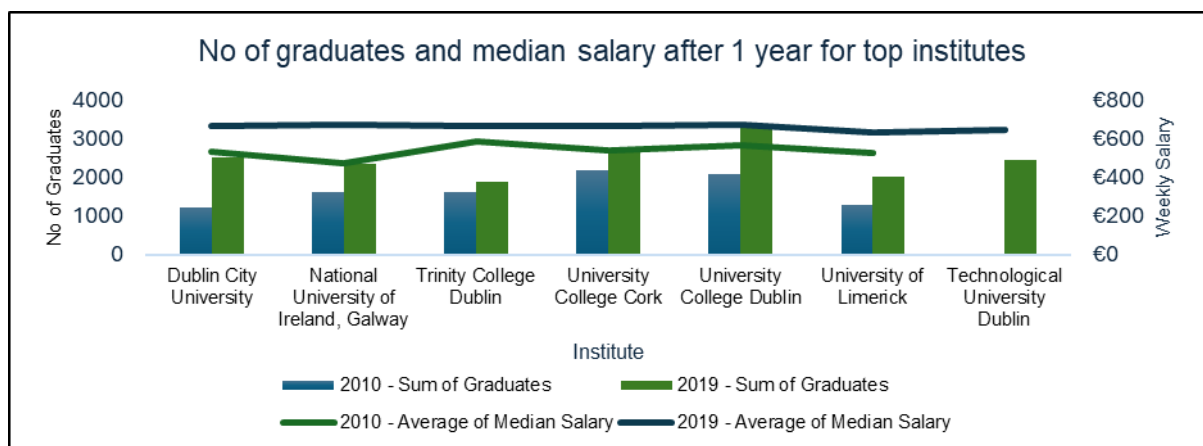


An analysis of gender-based salary outcomes through the above graphs indicate a **€14 weekly gap in favour of males** more per week than females, except in "*Education*" and "*Social-Sciences-Journalism-Information*" field of studies where females earn marginally more.



Over the decade, fields such as "*Health&Welfare*" "*Education*", and "*Business-Administration&Law*" consistently **leads in starting salary**, while "*Services*" and "*Arts&Humanities*" remain at the lower end. Also, starting salaries in "*Health&Welfare*" and "*Education*", have declined since 2010, in stark contrast "*Business, Administration and Law*" and "*Information and Communication Technologies*" which saw a **131%**

increase. The salary versus years-since-graduation curve shows that salary growth stagnates after a few years as graduates' working experience increases.



For institutions within the top three tiers, salary differences have diminished over time as shown in graph, with the mean weekly salary now standing at €658, despite noticeable differences in the number of graduates. The top three institutes, based on average weekly salary for each field of study, are as follows:

Field of Study	Institute
Agriculture, Forestry, Fisheries and Veterinary	University College Cork
	University College Dublin
	Athlone IT
Arts and Humanities	St. Patrick's College
	Trinity College Dublin
	Mary Immaculate College
Business, Administration and Law	Trinity College Dublin
	University College Dublin
	Dublin City University
Education	Dundalk IT
	Waterford IT
	St Angela's College of Home Economics
Engineering, Manufacturing and Construction	Trinity College Dublin
	University of Limerick
	Dublin City University
Health and Welfare	Royal College of Surgeons
	Trinity College Dublin
	University College Dublin
Information and Communication Technologies	Trinity College Dublin
	University College Dublin
	National College of Art and Design
Natural Sciences, Mathematics and Statistics	Dublin City University
	Mary Immaculate College
	University of Limerick
Services	National University of Ireland, Galway
	University College Dublin
	Dublin Institute of Technology
Social Sciences, Journalism and Information	Trinity College Dublin
	University College Dublin
	University of Limerick

Inferential Statistics

One Mean Z-Test

Reforms in 2014 introduced a €30,000 annual salary threshold (€577 weekly) for General Employment Permits. A right tailed one-mean z-test at 1% significance level was conducted to evaluate if graduates are hired with jobs satisfying the cutoff threshold or not.



Test confirmed that data provided sufficient evidence (test statistics = 3.80) that graduates' **mean salaries** within two years of graduation **exceeded threshold** with statistical significance. These trends reinforce Ireland's reputation as an attractive destination for international students.

Two Mean Z-Test

Descriptive statistics revealed an overall €14 weekly salary gap between genders. To further investigate, a right-tailed two-mean z-test at 5% significance level was performed to assess the significance of this difference. Results of the test found no statistically significant evidence (test statistics = 0.23) of gender disparity in starting salaries suggesting that **salary distribution is fair**.

Correlation T-Test

Mean weekly salaries increase proportionally with NFQ levels, from €653 for Level 8 to €934 for Level-10 graduates. A right-tailed correlation t-test was conducted to prove this further at 1% significance level. Test results confirmed the positive correlation with sufficient evidence (test statistics = 4.86), highlighting the value of advanced qualification in salary trends.

Non – Pooled T –Test

Descriptive statistics noticed that the differences in salaries across institutional tiers have diminished over time, and to further investigate, right-tailed non-pooled t-test was conducted at 1% significance level for institutions within tier 1 and 2.

The test failed to provide sufficient evidence (test statistic = -0.05) to identify a difference in earnings between Tier 1 and Tier 2 institutions, which aligns with the findings from the descriptive analysis. The narrowing of salary differences suggests either a shift in employer perceptions or a reduction in the distinctiveness of institutions. Institutions should consider strategies to strengthen their unique offerings to sustain a competitive edge.

Conclusion

The findings offer a nuanced understanding of factors influencing employability, industry preferences, and salary trends among graduates in Ireland.

Key Insights:

1. Impact of Academic Performance:

- The effects of grades (H1, H21, H22, and H3) on job outcomes were negligible. Higher academic grades did not significantly increase the likelihood of employment, according to hypothesis tests conducted on both genders.
- There is a need for educational institutions to improve soft skills, internships, and job experience in their curricula because employers seem to appreciate these more than academic grades.

2. Institutional Tier Influence:

- An ANOVA test revealed significant differences in employment outcomes across institutional tiers, emphasizing disparities in resources and networks. This underlines the importance of institutional strategy in fostering employability. However, salary was independent of the tier as concluded from non-pooled t test.

3. Gender-Based Employment and Salary Disparities:

- Although descriptive data indicated that women had greater employment rates, inferential testing revealed no discernible gender disparities.
- There was little statistical support for notable differences in beginning pay, and gender salary discrepancies were negligible.

4. Industry Preferences and Field of Study:

- *Business, Administration, and Law* emerged as the most popular fields for both genders, with *Education* being female-dominated and *Professional/Scientific* roles male-dominated.
- Salary growth fluctuated by field, with *Business* and *ICT* demonstrating the largest rises over the course of the decade, while *Arts and Services* continued to be at the bottom.

5. Graduate Salaries:

- Median weekly salaries increased steadily, though at a modest pace. Higher NFQ levels were strongly correlated with increased earning potential.
- The salary threshold for general employment permits was consistently surpassed, reinforcing Ireland's position as a favourable destination for skilled graduates.

6. Employment Trends:

- Less than 50% of graduates secured employment within two years out of employed graduates within 10 years, pointing to potential barriers in workforce integration that need to be addressed through tailored policy interventions.



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Contribution

Three authors examined several facets of the graduate outcomes data for this report:

Manodhya Gangaprasad analysed "What Do Graduates Do?" with an emphasis on career advancement and employment results.

Sarah Thomas analysed the industries and employment sectors that graduates enter in her study "Where Do Graduates Work?"

Anju Alex analysed the salary distribution among graduates while working on the issue, "What Do Graduates Earn?"

To address important concerns regarding graduate outcomes, such as the connection between education, employment sectors, incomes, and career trajectories, each contributor has used a particular Excel workbook to investigate their assigned region. With this research, we hope to offer insightful information to companies, educational institutions, and potential students in Ireland.